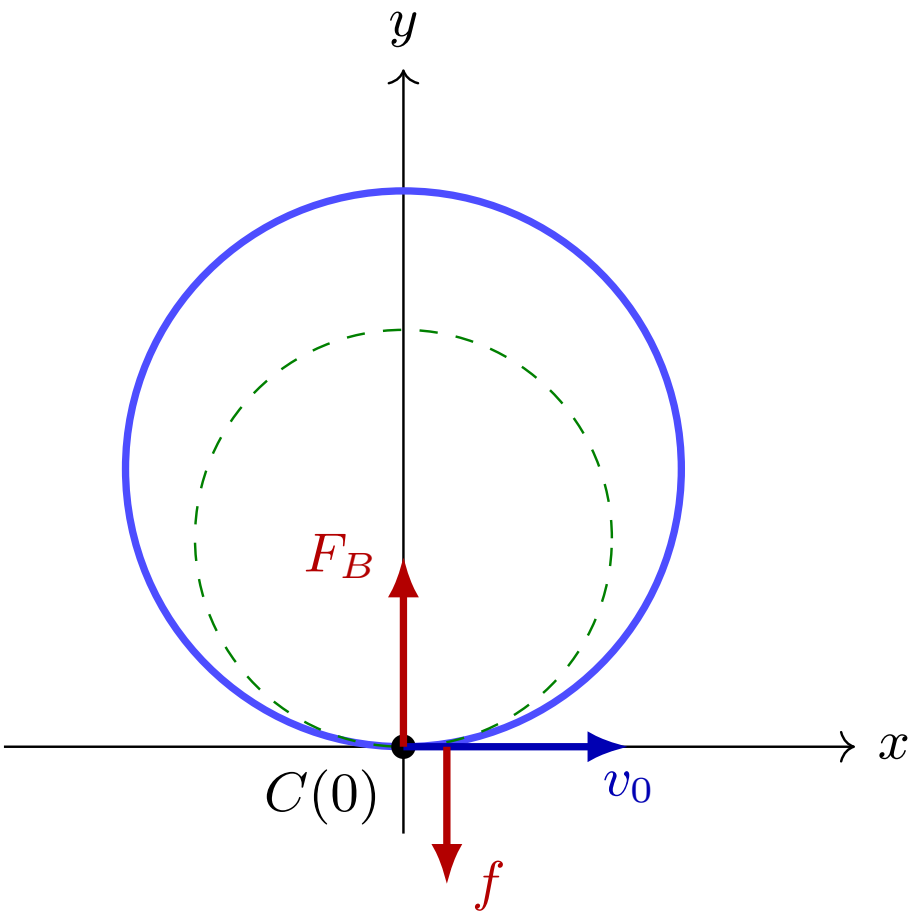
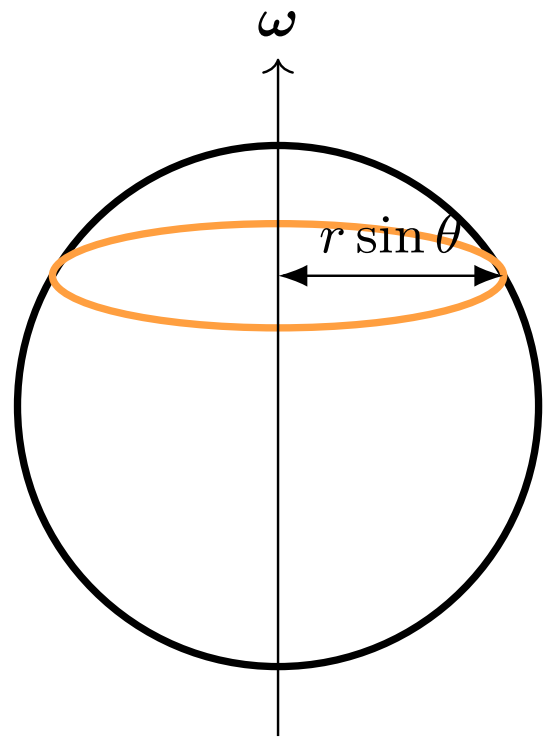


蓝：中心电荷 R_c

虚线：表面电荷 $R_c' = 3R_c/4$



$$\mathrm{d}\mu_m = \mathrm{d}I\,\pi r^2 \sin^2\theta$$

$$\boldsymbol{\mu}_m = (qr^2/3)\boldsymbol{\omega}$$